Neighbours of a square in a paired selection

Human Compression project: a component of the L044 recovery

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle$ be a transitive permutation group on Ω, generated by involutions with $[\rho_i, \rho_j] = 1$ when $|i - j| > 1$. An i-edge is eligible if its endpoints belong to different orbits of $G_i = \langle \rho_j : j \neq i \rangle$. A paired selection chooses two eligible edges of each colour on the full vertex set. Throughout this component, $n = |\Omega| = 2r$ and there is a connected paired selection Q whose unique cycle is an alternating square. The acquired normal-form and equality-connectedness arguments supply precisely this hypothesis in the remaining even case. Call a full-graph square *admissible* if it is the unique cycle of such a connected selection.

We will choose the selection separately for each admissible square. This qualification matters: we do not assert that every selection already contains all the neighbours of its square. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 4. The extremal choice and the explicit eight-vertex argument below replace the brief diagram arguments used there. No intersection property or parity hypothesis is needed for the statements in this component.

Lemma 1 (The four root trees)**.** *Let S be the unique square of Q, with colours a, b. Delete its four edges, and denote by T_p the tree rooted at each $p \in S$. The two components of Q obtained by deleting its a-edges are exactly the two actual G_a-orbits; the same holds for b. The four intersections of these orbit partitions are the vertex sets T_p. Consequently every ρ_k, $k \notin \{a, b\}$, preserves each T_p.*

Every eligible k-edge, $k \notin \{a, b\}$, lies in a tree containing a selected k-edge. If a selected k-edge meets a square vertex p and ρ_k commutes with ρ_a, then the other selected k-edge lies in $T_{p^{\rho_a}}$. In particular the colour of a selected edge leaving the square is adjacent to at least one of a, b in the integer path of labels.

Proof. Deleting the opposite a-edges leaves two connected components. Each is contained in an actual G_a-orbit, while eligibility of either deleted edge says that those orbits are distinct. As the components exhaust Ω, they are exactly the two orbits. The assertion for b follows in the same way. The two partitions cross at the four square vertices, and every remaining selected edge lies in its root tree. This proves the intersection statement and invariance under every other mark.

If a root tree has no selected k-edge, its tree connects every pair of its vertices without k. It therefore contains no eligible k-edge. Transport the given eligible edge at p by ρ_a. Commutation supplies another actual k-edge at p^{ρ_a}; membership $\rho_a \in G_k$ preserves eligibility. Its different root tree must contain the second selected k-edge. If k commuted with both square colours, transport to all four square vertices would require selected k-edges in four distinct trees. This is impossible, and proves the last assertion. □

Lemma 2 (A suitable selection at a fixed square)**.** *For each admissible square S, a connected paired selection with unique cycle S can be chosen so that every full-graph edge incident with a vertex of S is selected. In this selection, if an edge of colour c leaves S and c commutes with a square colour a, both parallel c-edges of that commuting square are selected.*

Proof. Among connected paired selections with unique cycle S, maximise the number of selected edges incident with S. This is a maximum over a finite nonempty set.

First let pu be a selected c-edge leaving S, with c commuting with a. Put $q = p^{\rho_a}$ and $w = u^{\rho_a} = q^{\rho_c}$. The vertices u, w lie outside S, since the a-matching already pairs its four

vertices. They lie in T_p, T_q, respectively. The eligible parallel edge qw forces the tree path from q to w to contain a selected c-edge. This is exactly the other selected c-edge, since pu lies in the different tree T_p. If qw were unselected, remove that other c-edge and insert qw. Removing the edge cuts T_q into the two parts containing q, w; the new edge reconnects them. Thus the selection remains connected with unique cycle S. The removed edge did not meet S: the only possible root is q, whose actual c-edge is qw. The number being maximised increases by one, a contradiction. This proves the parallel-edge assertion.

Now suppose an actual k-edge qv at $q \in S$ is unselected. Its colour is not a square colour, and its other endpoint belongs to $T_q \setminus S$. The first colour ℓ on the tree path from q to v differs from k, because the actual k-edge at q is qv. The path together with qv is a simple full-graph cycle. An eligible selected edge cannot be its only edge of that colour: the remainder of the cycle would connect its endpoints inside an omission orbit. Hence both selected ℓ-edges lie on this path, in T_q. By Lemma 1, ℓ commutes with neither a nor b. Therefore $|a - b| = 2$ and ℓ is their common middle index.

Since $k \neq \ell$, it commutes with at least one square colour, say a. Put $q' = q^{\rho_a}$ and $v' = v^{\rho_a}$; the latter is outside S. The actual parallel k-edge $q'v'$ lies in $T_{q'}$. It too is unselected, since otherwise the first part of this proof, applied in the opposite direction, would make qv selected. Applying the preceding path argument in $T_{q'}$ forces both selected edges of the common middle colour ℓ to lie there. They already lie in T_q, a contradiction. □

Lemma 3 (Two commuting branches)**.** *In a root tree T_p, suppose two selected edges pu, pv have different commuting colours c, d, with exactly one selected edge of each of these colours in T_p. This is impossible.*

Proof. Both marks preserve T_p by Lemma 1. Commutation supplies $w = u^{\rho_d} = v^{\rho_c} \in T_p$ and transports eligibility to the edges uw of colour d and vw of colour c. The four vertices are distinct: a collision would make the initial two differently coloured edges coincide, contrary to eligibility. In the tree T_p, deletion of pu leaves p, v on the same side. Since vw is eligible and pu is the only selected c-edge in the tree, w must be on the u side. Likewise deletion of pv forces w to be on the v side. The two branches at p are disjoint, a contradiction. □

Lemma 4 (Moving to a neighbouring square)**.** *If an actual edge of colour c leaves an admissible square S of colours a, b and c commutes with a, then the commuting square of colours a, c sharing that a-edge with S is admissible.*

Proof. Use the selection of Lemma 2, and write the two parallel c-edges as pu, qw, where $q = p^{\rho_a}$ and $w = u^{\rho_a}$. Replace the selected a-edge of S opposite pq by uw. The latter is eligible by transport by ρ_c. Deleting an edge of the unique cycle leaves a connected tree; adding uw makes its unique cycle precisely $pquwp$, with colours a, c. Both new vertices are outside S and in its different root trees. The edge counts and eligibility are unchanged, proving admissibility. □

Theorem 5 (The degree at a square)**.** *If $n > 8$, every vertex of every admissible square has degree at most three in the full permutation graph. Degrees here count coloured edges; Lemma 2 also excludes multiple edges incident with such a vertex.*

Proof. Fix an admissible square with colours a, b, and choose its selection as in Lemma 2. Suppose a vertex p has two further incident colours c, d. Each is adjacent to at least one square colour. If neither is the common middle of square labels at distance two, each commutes with at least one square colour. Lemma 1 then puts the other selected edge of each colour in a different root tree. Thus c, d cannot commute, by Lemma 3.

Two distinct integers outside $\{a, b\}$, each adjacent to at least one of a, b, can themselves be adjacent only as the two inner indices of four consecutive integers. To check this, if both are neighbours of the same square index their distance is two. Otherwise write $c = a \pm 1, d = b \pm 1$

with $a < b$; the equation $|c-d| = 1$, excluding $c = b, d = a$, leaves $b = a+3, c = a+1, d = a+2$. After reversal, the square and its attachments therefore have labels

$$a, \quad k = a+1, \quad h = a+2, \quad b = a+3,$$

with square colours a, b and attachments k, h at p.

The exceptional possibility in the preceding paragraph is a square with colours $a, a+2$, with one attachment of middle colour $a+1$. Its other attachment has colour $a-1$ or $a+3$. Reverse all labels if needed to make it $a+3$. Move the square along this latter attachment using Lemma 4. At the same vertex p, the new admissible square has outer colours $a, a+3$ and retains the actual attachments $a+1, a+2$. The selection can be chosen afresh by Lemma 2. Thus it suffices in every case to exclude the four-consecutive-label configuration just displayed.

Write the square vertices as $p, q = p^{\rho_a}, s = p^{\rho_b}, t = p^{\rho_a\rho_b}$. The parallel-edge assertion gives selected k-edges pu, su' and selected h-edges pv, qv'. These, together with the square, use both selected edges of all four consecutive colours. The eight vertices

$$p, q, s, t, u, u', v, v'$$

are distinct. Indeed u, v are different neighbours of p in T_p, and u', v' lie in T_s, T_q, respectively; all four are outside the square.

We show that every colour outside these four fixes the eight vertices. If such a colour moves a square vertex, it must be $a-1$ or $b+1$, by Lemmas 1 and 2. For $d = a-1$, an edge at p or s would commute with the existing k-edge there. Both colours commute with b, so each has just one selected edge in that root tree, contrary to Lemma 3. An edge at q would commute with the existing h-edge; the other h-edge lies at p and the other d-edge would lie at t, giving the same contradiction. Finally d commutes with b, so fixing q forces it to fix $t = q^{\rho_b}$. This excludes d at every square vertex.

For $e = b+1$, edges at p or q conflict in the same way with the h-edges, since both colours commute with a. An edge at s conflicts with the k-edge there, whose other edge lies at p, whereas the other e-edge would lie at t. Commutation with a then transfers fixation from s to t. This excludes e as well. These arguments simply omit a label if it is outside $0, \ldots, r-1$.

Every colour outside the four consecutive ones commutes with both k and h, whose only adjacent indices are within the four. Since it fixes p, q, s, it therefore fixes $u = p^{\rho_k}$, $u' = s^{\rho_k}$, $v = p^{\rho_h}$ and $v' = q^{\rho_h}$. Thus no selected edge of an outside colour meets the eight vertices. Nor can another selected edge of one of the four internal colours leave them: both selected edges of each are already among the eight edges displayed. They form an entire connected component of Q on eight vertices. Connectedness of Q gives $n = 8$, contradicting the hypothesis. □

Scope and remaining obligation. This component proves local normalisation, admissible square transport and the actual degree bound. It does not yet classify the selected graph as a ladder or prove that G preserves pairs. Those global steps and the separate transitive-imprimitive omission case remain necessary for the principal L044 theorem. No whole original, open question, Lean theorem or compression measurement is credited here.